

Game Theory – 1

Impartial games: P-positions and the Sprague–Grundy theory

Homework

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Theoretical minimum. A game is impartial if both players have the same moves from every position, finite if no play is infinite, and played under the normal convention: whoever cannot move loses. By backward induction the positions split into P-positions (the player to move loses) and N-positions; a terminal position is a P-position, a position is an N-position exactly when some move leads to a P-position, and the winning strategy is “always move to a P-position”. In Nim the P-positions are those with $a_1 \oplus \dots \oplus a_k = 0$, where $\oplus$ is bitwise exclusive or. The Grundy value is $g(x) = \text{mex}\{g(y) : x \rightarrow y\}$, where mex is the least non-negative integer not occurring; $g(x) = 0$ characterises P-positions, a Nim heap of n has $g = n$, and by the Sprague–Grundy theorem the Grundy value of a sum of games is the $\oplus$ of the Grundy values. Hence every finite impartial game is a disguised Nim heap. For subtraction games with a finite move set the Grundy sequence is eventually periodic. None of this survives if the two players have different moves, or if the convention is *misère*.

1. Let k and n be integers with $1 \leq k \leq n$. Alice and Bob play with k pegs in a line of n holes; at the start the pegs occupy the k leftmost holes. A legal move takes a single peg to any vacant hole further to the right. The players alternate, Alice first, and the game ends when the pegs occupy the k rightmost holes, so that the player to move loses. For which n and k does Alice have a winning strategy? (Putnam 2020)
2. Let $n \geq 1$ be odd. Alice and Bob alternate, Alice first, on a line of n spaces, initially empty. At each turn a player either places a stone in

an empty space, or removes a stone from a non-empty space s , places a stone in the nearest empty space to the left of s (if any) and a stone in the nearest empty space to the right of s (if any). The player who fills the last empty space wins. Determine the winner. (Putnam 2013)

3. Wythoff's game: two heaps, and a move removes any positive number of tokens from one heap, or the same positive number from both. Prove that the P-positions are exactly the pairs $(\lfloor n\varphi \rfloor, \lfloor n\varphi^2 \rfloor)$, $n \geq 0$, where $\varphi = \frac{1+\sqrt{5}}{2}$; use Beatty's theorem that $\lfloor n\varphi \rfloor$ and $\lfloor n\varphi^2 \rfloor$ together list every positive integer exactly once.
4. Fibonacci nim: one heap of n tokens; the first move may take any number of tokens except the whole heap, and afterwards a player may take at most twice what the opponent has just taken. Prove that the first player loses exactly when n is a Fibonacci number, and explain the role of the Zeckendorf representation.
5. Two players play the following game. Let $n > 1$ be a fixed integer. Starting from $k = 2$, each player may replace the current number k either by $k + 1$ or by $2k$. The player forced to write a number greater than n loses. For which n does the first player have a winning strategy? (VJIMC 2006)
6. Alice and Bob play with a string of n digits, each restricted to 0, 1 or 2; initially all digits are 0. A legal move adds or subtracts 1 from one digit so as to create a string that has not appeared before. A player with no legal move loses. Alice goes first. For each $n \geq 1$ determine the winner. (Putnam 2025)
7. Alice and Bob play with a chocolate bar in the shape of a 20×23 rectangle. Alice breaks off a part along a groove and discards it, then Bob breaks off a part of what remains and discards it, and so on until a single 1×1 piece is left; the player who leaves the opponent with that single piece wins. Which player has a winning strategy? (OMOUS 2023)
8. Euclid's game: a position is a pair of positive integers, and a move

replaces the larger by the difference of the larger and any positive multiple of the smaller, provided the result is non-negative; a player who produces a zero wins. Prove that the position (a, b) with $a < b$ is an N-position if and only if $b/a \geq \varphi$, and explain where the golden ratio comes from.

9. Turning Turtles: a row of n coins, each heads or tails; a move turns over one coin from heads to tails and may additionally turn over any one coin to its left, in either direction. Prove that the Grundy value of the position equals the $\oplus$ of the positions of the coins showing heads, so that the game is Nim in disguise.
10. Dawson's Kayles is played on a row of n squares: a move removes two adjacent squares, splitting the row into two independent rows. Compute the Grundy values for $n \leq 20$ by hand, using the Sprague–Grundy theorem to treat a split position as a sum, and check that no period is visible yet (the true period is 34, with finitely many exceptions). Then state precisely what is known in general: for a finite subtraction set the Grundy sequence is always eventually periodic, while for games of this type the periodicity of the Grundy sequence is an open problem.